1. Introduction

This task focuses on performing Exploratory Data Analysis (EDA) on the Titanic dataset.
The objective is to understand the dataset structure, detect patterns, and visualize relationships between different features.

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2. Importing Required Libraries

```
import pandas as pd

import numpy as np

import seaborn as sns

import matplotlib.pyplot as plt


%matplotlib inline

sns.set(style="whitegrid")

print("Libraries imported successfully")
```

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3. Load Dataset

```
df = pd.read_csv("titanic.csv")

df.head()
```

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4. Dataset Overview

4.1 Dataset Info

```
df.info()
```

4.2 Summary Statistics

```
df.describe()
```

4.3 Missing Values

```
df.isnull().sum()
```

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5. Visualizations

5.1 Histogram (Age & Fare)

```
df[['Age', 'Fare']].hist(figsize=(10,4), bins=20)
plt.suptitle("Distribution of Age & Fare")
plt.show()
```

5.2 Boxplots (Age & Fare)

```
plt.figure(figsize=(10,4))

plt.subplot(1,2,1)
sns.boxplot(y=df['Age'])
plt.title("Age Boxplot")

plt.subplot(1,2,2)
sns.boxplot(y=df['Fare'])
plt.title("Fare Boxplot")

plt.tight_layout()
plt.show()
```

5.3 Correlation Heatmap

```
corr = df.corr(numeric_only=True)
```

```
plt.figure(figsize=(10,6))

sns.heatmap(corr, annot=True, linewidths=0.5)

plt.title("Correlation Heatmap")

plt.show()
```

5.4 Pairplot (Sample)

```
sample_df = df.sample(150, random_state=42)

sns.pairplot(sample_df[['Survived','Pclass','Age','Fare']], diag_kind="kde")

plt.show()
```

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6. Saving Processed Dataset

```
df.to_csv("eda_used_titanic.csv", index=False)

print("eda_used_titanic.csv saved")
```

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7. Observations

✔ Age Distribution:

Most passengers fall between 20–40 years.

✔ Fare Distribution:

Highly skewed; few passengers paid significantly higher fares.

✔ Survival Patterns:

- Females survived more than males.
- 1st class passengers had higher survival rates.

✔ Correlation:

Fare and Pclass show clear negative correlation (higher class → higher fare).